

ELEN 4193 Flicker Fusion Lab

Songhang Li

February 2026

1 Flicker Fusion and the Persistence of Vision

1.1

Under normal light conditions, the threshold frequency is about 50-60 Hz

1.2

Yes, there is a slight difference upon the threshold frequency between the colors.
Red: 50 Hz. Green: 60 Hz. Blue: 53 Hz.

1.3

Yes, peripheral vision has a higher flicker threshold frequency. Under the same frequency as in the previous part where I felt continuous light, I can observe noticeable flicker with peripheral vision.

1.4

If the display frequency is lower than the eyes' flicker threshold frequency, then the content will look more flicking/ flashing, rather than a smooth image. Additionally, higher refresh rates reduce motion blur, especially important for gaming and fast-moving content.

1.5

If my threshold is higher, the content shown on the displays would look more blurred and flicking. If my threshold is lower, the content would look more smooth and continuous.

The higher the frequency, the higher the refresh rate. The high frequency is used to accommodate more users that have a higher threshold frequency. In some scenario like competitive gaming, high refresh rate displays are required, because they have lower lag.

2 Color Mixing

2.1

As the blue has higher intensity than other 2 colors, the mixing result looks more blue or purple. This may cause by manufacture tolerance or the characteristic of luminance material for different colors.

2.2

Yes, the ambient conditions affect the color perception significantly. Compared with the darker indoor lighting, the outdoor sunlight wash out the LED light making it less saturated.

Yes it should because it's fundamental to human color perception. Our visual system constantly adapt to viewing conditions.

2.3

Yes. Turning off the green and leaving red and blue on by digital output pins produced magenta. It's dominated by the blue.

2.4

Yes, by adding another 220 Ω resistor in series with the one with the blue, it is closer to the desired magenta.

2.5

Yes, the analogWrite method approximates magenta better.

```
analogWrite(R, 255); // (100%)  
analogWrite(B, 128); // (50% duty cycle)
```

2.6

They are not the same. The analog method controls the color intensity by limiting the current through LED, while the PWM method controls intensity by adjusting the frequency of different colors. I would choose the PWM method because of its precise control of color mixing. And it also require less component.

A Appendix

A.1 Frequency

```
const int R = 10;  
const int B = 11;  
const int G = 12;
```

```

const int f = 53; // Herz
unsigned long halfPeriod; // half period in ms
void setup() {
    pinMode(R, OUTPUT);
    pinMode(B, OUTPUT);
    pinMode(G, OUTPUT);

    digitalWrite(R, LOW);
    digitalWrite(B, HIGH);
    digitalWrite(G, LOW);

    halfPeriod = 1000 / (2 * f); // ms
}

void loop() {
    digitalWrite(B, LOW);
    delay(halfPeriod);
    digitalWrite(B, HIGH);
    delay(halfPeriod);
}

```

A.2 Analog Control

```

const int R = 10;
const int B = 11;
const int G = 12;

void setup() {
    pinMode(R, OUTPUT);
    pinMode(B, OUTPUT);
    pinMode(G, OUTPUT);

    digitalWrite(G, LOW);
    analogWrite(R, 255); // ON (100%)
    // Blue reduced brightness (50% duty cycle)
    analogWrite(B, 128); // 50%
}
void loop() {}

```

B Circuit

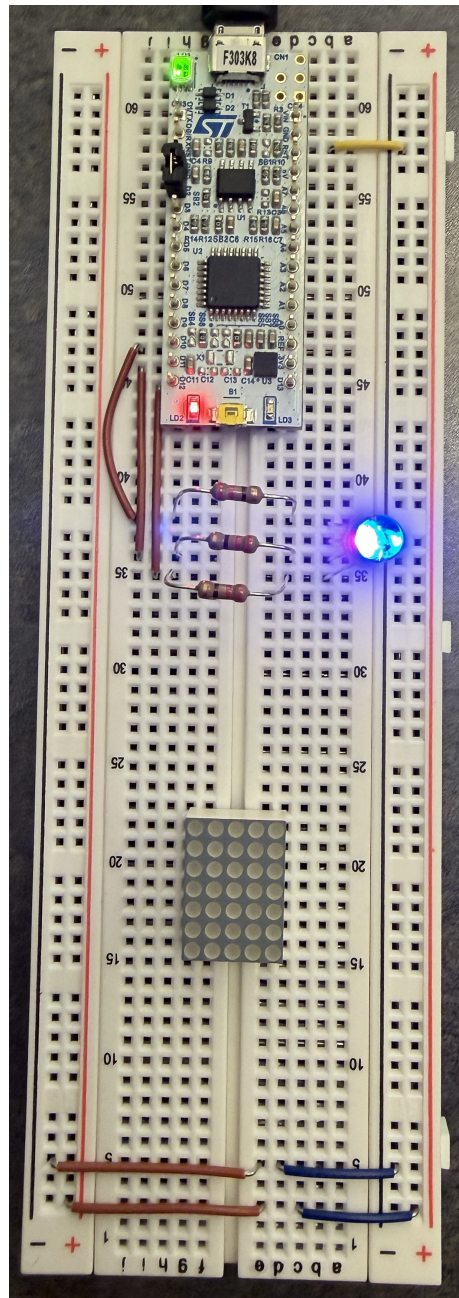


Figure 1: Circuit